



Summer Term 2026

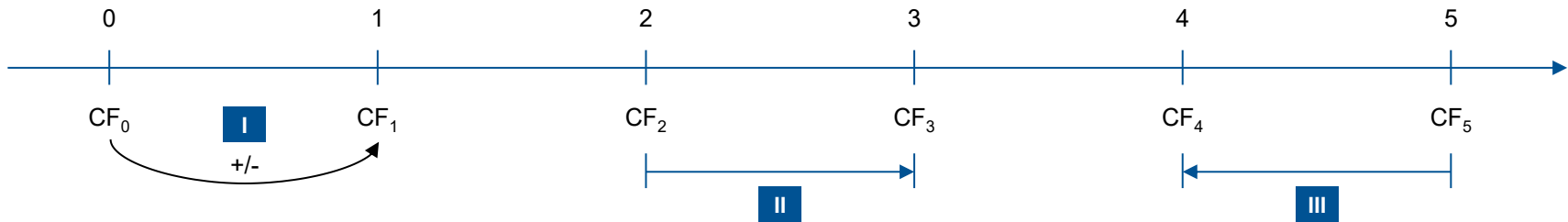
Investment and Financial Management
Exercise on Mathematical Basics

Session 03: Annuities

Center for Digital Transformation

Technische Universität München (TUM), Campus Heilbronn

Core Principle I: The Time Value of Money



I

It is only possible to **compare** or **combine** values that refer to the **same point in time**.

II

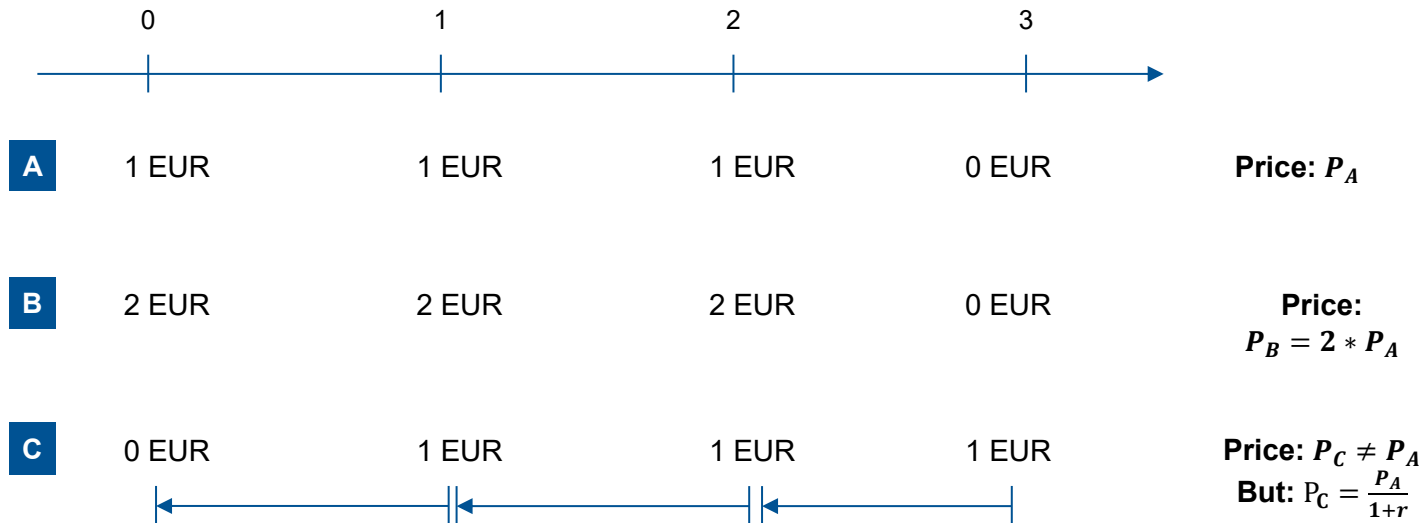
To move a cash flow forward in time, you must **compound** it (Future Value).

III

To move a cash flow back in time, we must **discount** it (Present Value).

The interest rate is applied to convert time values from one period to another!

Core Principle II: No Arbitrage in perfect financial markets



Same as two times contract A

Payments of contract A are only lagged by one period

I

Two securities that provide the **same future cash flows** and have the **same level of risk** must sell for the **same price**.
There is “no way to make money out of nothing”.

Core Principle II: No Arbitrage in perfect financial markets Example

	0	1
		
Security A	$P_0^A = 5 \text{ EUR}$	<i>Good state:</i> $CF_1^A = 7 \text{ EUR}$ <i>Bad state:</i> $CF_1^A = 5 \text{ EUR}$
Security B	$P_0^B = 11 \text{ EUR}$	<i>Good state:</i> $CF_1^B = 14 \text{ EUR}$ <i>Bad state:</i> $CF_1^B = 10 \text{ EUR}$

Investment Strategy: Buy two times security A and sell security B!

Strategy	$CF_0 = (-5 \text{ EUR}) * 2 + 11 = \mathbf{1 \text{ EUR}}$	<i>Good state:</i> $CF_1 = (2 * 7 \text{ EUR}) - 14 \text{ EUR} = 0 \text{ EUR}$ <i>Bad state:</i> $CF_1 = (2 * 5 \text{ EUR}) - 10 \text{ EUR} = 0 \text{ EUR}$
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Arbitrage Opportunity: No matter which state will occur, an investment strategy will make a certain profit today!

- 2.1 Introduction**
- 2.2 Application of annuity calculation**
- 2.3 Constant annuities**
 - 2.3.1 Yearly annuities**
 - 2.3.2 Intra-year annuities**
- 2.4 Varying annuities**
 - 2.4.1 Arithmetic progression**
 - 2.4.2 Geometric progression**

2.1 Introduction

2.2 Application of annuity calculation

2.3 Constant annuities

2.3.1 Yearly annuities

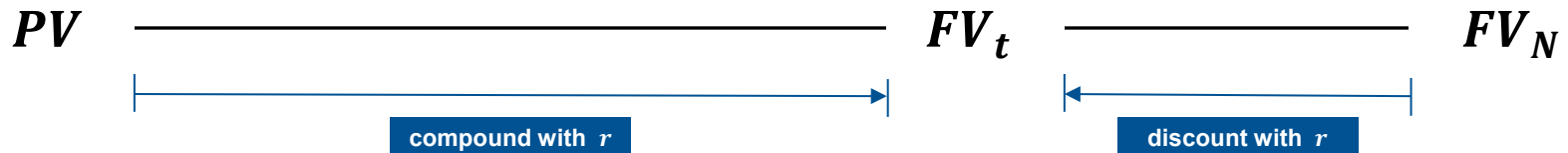
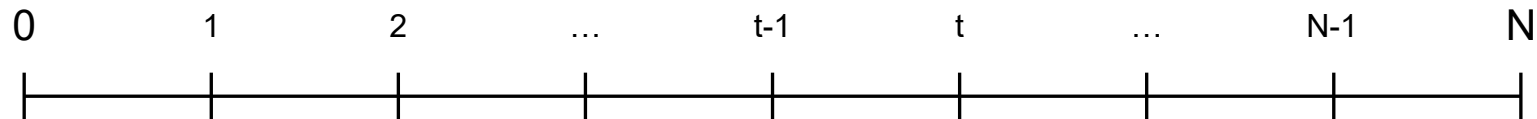
2.3.2 Intra-year annuities

2.4 Varying annuities

2.4.1 Arithmetic progression

2.4.2 Geometric progression

Annuity



Terms

C Periodically annuity payment
 N Investment period
 r Interest rate or discount rate
 q Discount factor ($1 + r$)

FV Future value of an annuity
 PV Present value of an annuity
 FVF Future value factor of an annuity
 PVF Present value factor of an annuity

Annuity-due vs. Annuity-immediate

Annuity-due

- Payments are made at the **beginning** of the corresponding time interval
- First payment at t_0 , last payment at t_{N-1}

Annuity-immediate

- Payments are made at the **end** of the corresponding time interval
- First payment at t_1 , last payment at t_N

No-Arbitrage Principle leads to:

$$(1): C_{due} * (1 + r) = C_{imm}$$

$$(2): C_{due} * q = C_{imm}$$

Payments are only
lagged by one interest
period

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When do we need to apply annuity calculation in real life?

- **Pay-in over several periods, one final payment**
 - Saving plan
 - Life insurance
 - Building loan contract
- **Pay-in over several periods, multiple final payments**
 - Retirement pension

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Saving (accumulation) phase

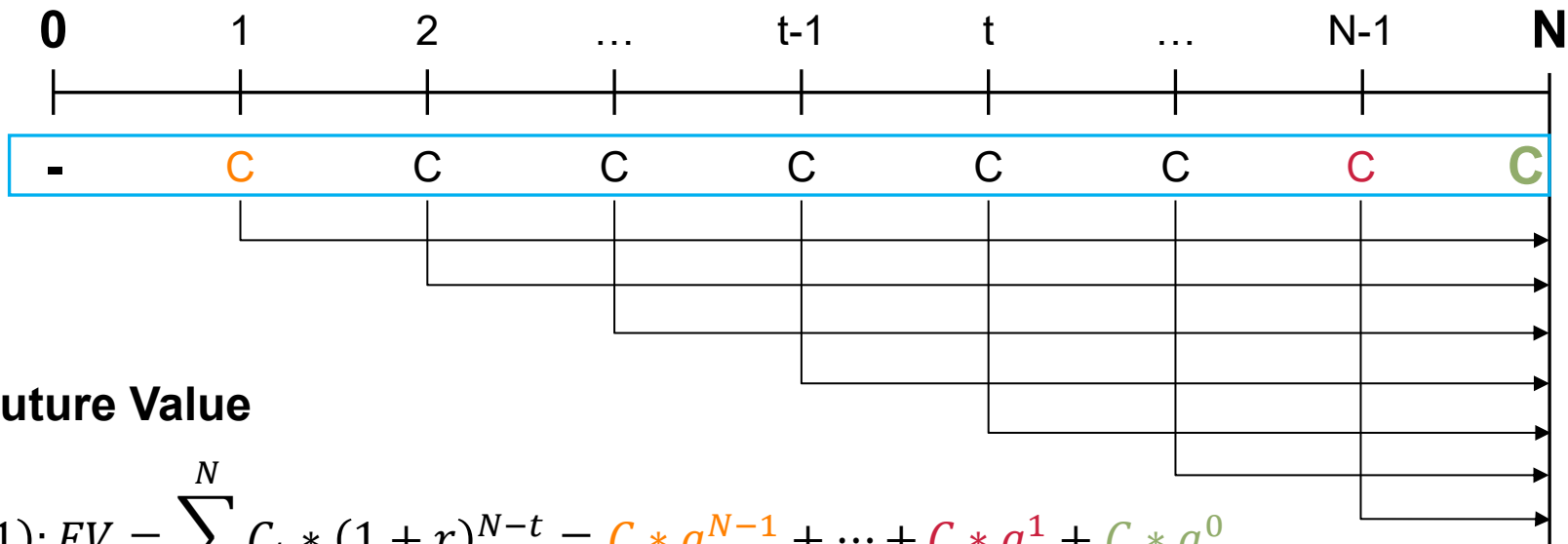
Regular payments of a constant amount C

- at the beginning of the investment interval (**due**)
- at the end of the investment interval (**immediate**)

0	1	2	...	t-1	t	...	N-1	N
C	C	C	C	C	C	C	C	-
-	C	C	C	C	C	C	C	C

Assumption: Due to complexity issues, we assume here that payment period and interest period are equal.

Future Value with annuities-immediate



Future Value

$$(1): FV = \sum_{t=1}^N C_t * (1 + r)^{N-t} = C * q^{N-1} + \dots + C * q^1 + C * q^0$$

$$(2): FV = C * \frac{q^N - 1}{q - 1} = C * FVF \text{ with } \frac{q^N - 1}{q - 1} = FVF$$

Future Value with annuities-immediate

Why does this hold?

$$(1): FV = C * \frac{q^N - 1}{q - 1} = C * FVF$$

Future Value Calculation using the principle of a **geometric series**:

$$(2): FV = \sum_{t=0}^N C_t * (1 + r)^{N-t} = C * (1 + q + q^2 + q^3 + \dots + q^{N-1})$$

$$(3): q * FV = C * (q + q^2 + q^3 + \dots + q^N)$$

$$(2) - (3): FV * (1 - q) = C * (1 - q^N)$$

$$(4): FV = C * \frac{(1 - q^N)}{(1 - q)} = C * \frac{(q^N - 1)}{(q - 1)}$$

(A.1) Building Loan Contract

For a building loan contract, a student's grandmother has paid 2,500 EUR at the end of each year on an account for her grandson. Interest is paid at the end of each year, the rate equals 3% with compounding interest.

(a) What will be the account balance after 30 years?

$$(1): FV = C * \frac{q^N - 1}{q - 1} = 2,500 \text{ EUR} * \frac{1.03^{30} - 1}{1.03 - 1} = \mathbf{118,938.54 \text{ EUR}}$$

(b) How much does she need to pay if the account balance should equal 200,000 EUR after 30 years?

$$(2): FV = C * \frac{q^N - 1}{q - 1}; 200,000 \text{ EUR} = C * \frac{1.03^{30} - 1}{1.03 - 1}$$

$$(3): C = FV * \frac{q - 1}{q^N - 1} = 200,000 \text{ EUR} * \frac{1.03 - 1}{1.03^{30} - 1} = \mathbf{4,203.85 \text{ EUR}}$$

(A.1) Building Loan Contract

(c) If she remains at 2,500 EUR per year, when will the account balance reach 100,000 EUR?

A = 40 years

B = 27 years

C = 20 years

D = 17 years

$$(F): FV = C * \frac{q^N - 1}{q - 1}$$

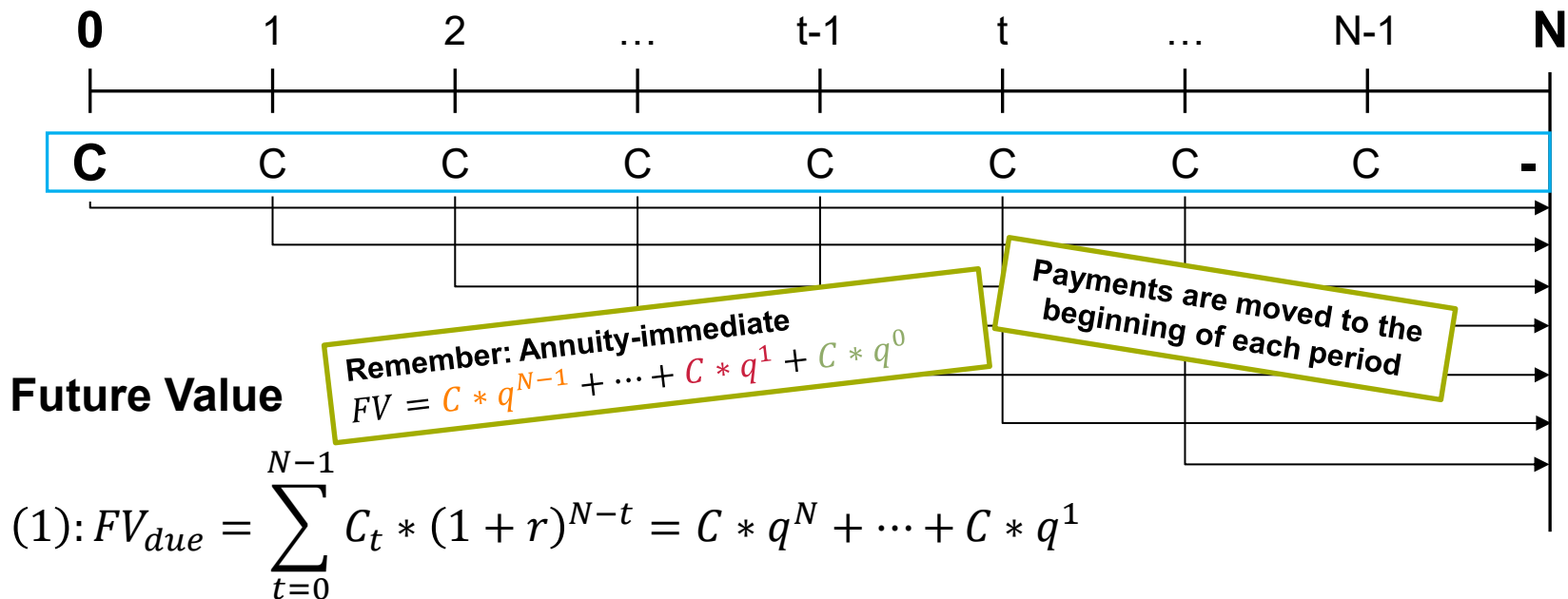
$$(1): q^N = \frac{C + FV(q - 1)}{C} \Leftrightarrow N = \frac{\ln\left(1 + \frac{q - 1}{C} * FV\right)}{\ln(q)} = \frac{\ln\left(1 + \frac{1.03 - 1}{2,500 \text{ EUR}} * 100,000\right)}{\ln(1.03)}$$

$$(2): N = 26.67 \text{ years}$$

(d) Would the future value change if the grandmother would pay at the beginning of each year starting just when the grandson is born?

We need to look at the concept of annuity-due!

Future Value with annuities-due



(2): $FV_{due} = C * \boxed{q} * \frac{q^N - 1}{q - 1} = C * FVF_{due}$

(A.1) Building Loan Contract

(d) Would the future value change if the grandmother would pay at the beginning of each year starting just when the grandson is born?

$$(F): FV_{due} = C * q * \frac{q^N - 1}{q - 1}$$

$$(1): FV_{due} = 2,500 \text{ EUR} * 1.03 * \frac{1.03^{30} - 1}{1.03 - 1} = 122,506.70 \text{ EUR} > 118,938.54 \text{ EUR}$$

Change comes from investing at the beginning of each period instead of at the end!

(A.1) Building Loan Contract

(e) If the account balance should equal 200,000 after 30 years, what would need to be the yearly payment if we consider annuities-due?

Please assume the following values: $r = 0.03$.

$$A = 2,831.23 \text{ EUR}$$

$$B = 3,476.25 \text{ EUR}$$

$$C = 4,081.41 \text{ EUR}$$

$$D = 4,203.85 \text{ EUR}$$

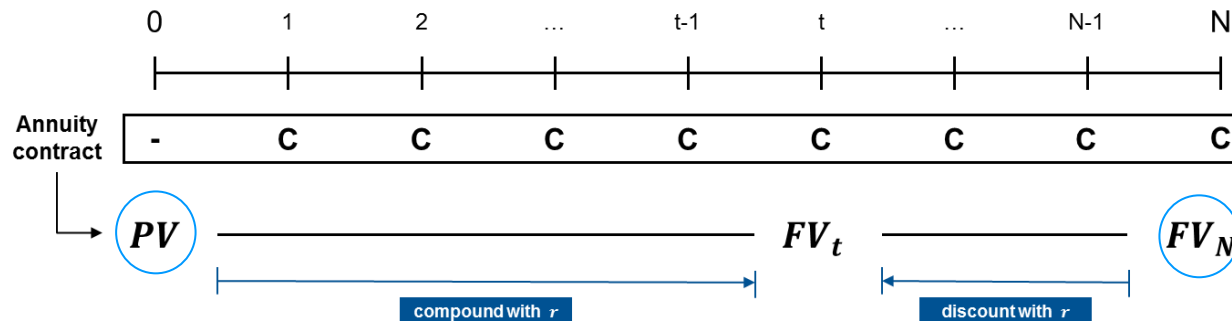
$$(F): FV_{due} = C * q * \frac{q^N - 1}{q - 1}; C = FV_{due} * \frac{1}{q} * \frac{q - 1}{q^N - 1}$$

$$(1): C = 200,000 \text{ EUR} * \frac{1}{1.03} * \frac{1.03 - 1}{1.03^{30} - 1} = 4,081.41 \text{ EUR} < 4,203.85 \text{ EUR}$$

$$\text{Remember – No arbitrage: } C_{due} * q = C \Rightarrow 4,081.41 \text{ EUR} * 1.03 = 4,203.85 \text{ EUR}$$

Present Value of an annuity

Idea: Here we follow the idea of the time value of money.



Remember:
Discounting the Future Value with r leads to the Present Value!

- Annuity-immediate

$$(1): PV = \frac{FV}{q^N} = C * \frac{FVF}{q^N} = C * \frac{1}{q^N} * \frac{q^N - 1}{q - 1} = C * \frac{q^N - 1}{q^{N+1} - q^N} = C * PVF$$

- Annuity-due

$$(2): PV_{due} = \frac{FV_{due}}{q^N} = C_{due} * \frac{FVF_{due}}{q^N} = C_{due} * \frac{q^{N+1} - q}{q^{N+1} - q^N} = C * PVF_{due}$$

(A.2) Mixed Exercises

Over a period of 20 years you invest an annual amount of 5,000 EUR at 4%. What will be your account balance if you consider...

(a) annuity-due?

(b) annuity-immediate?

$$FV = 154,846.01 \text{ EUR} \\ FV_{due} = 148,490.39 \text{ EUR}$$

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$$FV = 148,490.39 \text{ EUR} \\ FV_{due} = 154,846.01 \text{ EUR}$$

$$(1): FV_{due} = C_{due} * q * \frac{q^N - 1}{q - 1} = 5,000 \text{ EUR} * 1.04 * \frac{1.04^{20} - 1}{1.04 - 1} = 154,846.01 \text{ EUR}$$

$$(2): FV = C * \frac{q^N - 1}{q - 1} = 5,000 * \frac{1.04^{20} - 1}{1.04 - 1} = 148,890.39 \text{ EUR}$$

$$(3): \text{Alternative Solution: } FV = \frac{FV_{due}}{q}$$

(A.3) Mixed Exercises

Mr. Huber celebrates his 30th birthday on December 31, 2016. He decides to invest in a fixed-interest life insurance, which will pay off 250,000 EUR when he turns 65. The guaranteed interest rate will be 3.25% over the whole investment period. How much money need to be paid in at the end of each year?

$$(1): C = FV * \frac{q - 1}{q^N - 1} = 250,000 * \frac{1.0325 - 1}{1.0325^{35} - 1} = 3,938.37 \text{ EUR}$$

(A.4) Mixed Exercises

Mr. Huber regularly pays in 6,000 EUR at a rate of 6%. How many years does it take until the account balance is above 60,000 EUR? Please consider two contracts (annuity-due and annuity-immediate) and determine the time difference.

A = 0 years

B = 1 year

C = 2 years

D = 3 years

(F): $FV \equiv 60,000 \text{ EUR}$; $FV_{due} = 60,000 \text{ EUR} \rightarrow \text{Solve both equations for } N$

$$(1): N = \frac{\ln(1 + \frac{q - 1}{C_{due} * q} * FV_{due})}{\ln(q)} = \frac{\ln(1 + \frac{1.06 - 1}{6,000 \text{ EUR} * 1.06} * 60,000 \text{ EUR})}{\ln(1.06)} = 7.70$$

$$(2): N = \frac{\ln(1 + \frac{q - 1}{C} * FV)}{\ln(q)} = \frac{\ln(1 + \frac{1.06 - 1}{6,000 \text{ EUR}} * 60,000 \text{ EUR})}{\ln(1.06)} = 8.07$$

(3): 8 years < 9 years

Payout Phase

Idea: Regular annuity payments are financed by a saving account with an initial capital of $PV = K_0$, which is invested at a rate of r . At every interest payment date, the constant annuity C is paid ...

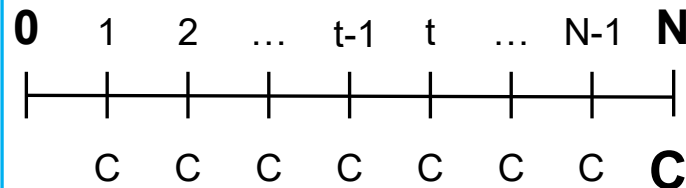
- at the beginning or ...
- at the end of the period.

Due to complexity we assume that interest payment date and payout date of the annuity are equal.

Payout Phase

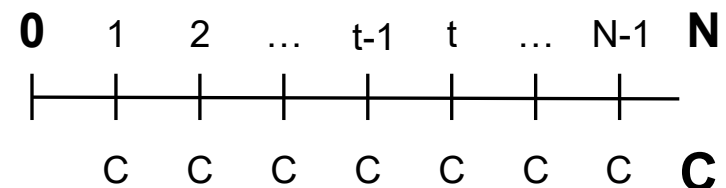
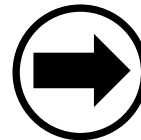


$$FV_{\text{Saving}} = PV_{\text{Payout}}$$



Saving Phase

e.g. employee saves on a pension account (saving annuity)



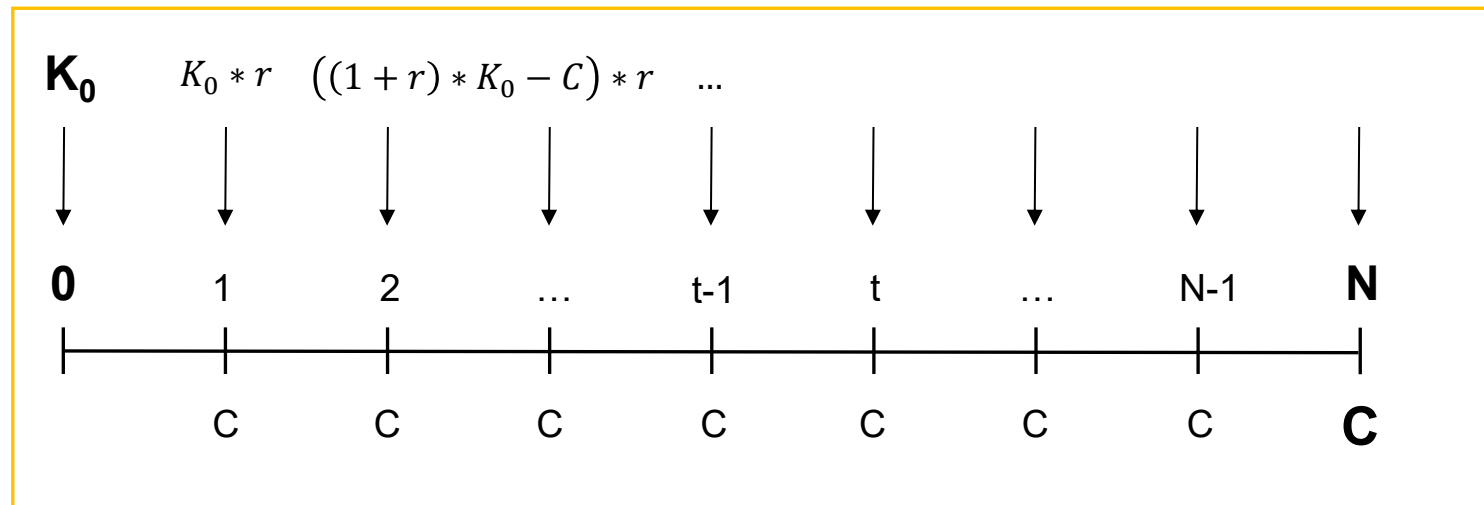
Payout Phase

e.g. employer pays regular amount to employee (payout annuity)

$$FV_{\text{Saving Annuity}} = PV_{\text{Payout Annuity}}$$

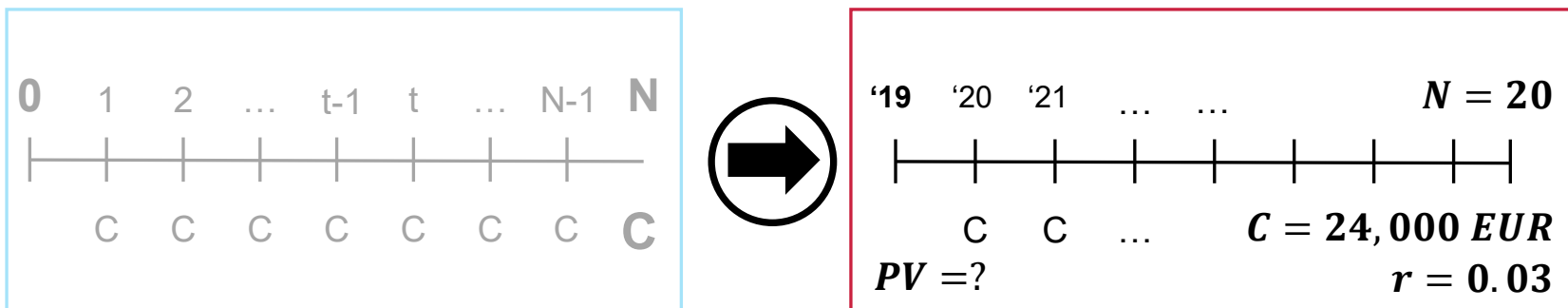
Payout Phase

- Interest is constantly paid on the initial capital K_0 . At the same time the amount is reduced by the repayment amount C at every payout date.
- In the same period, money flows in as interest earnings and out as repayment.



(A.5) Payout Phase

What needs to be the initial capital $K_0 = PV$ if there should be an annuity of 24,000 EUR over a time horizon of 20 years. Please consider an interest rate of 3% and annuity-immediate.



What is the contract worth today?

What would I have needed to save until today so that I will receive the determined cash flows over the next 20 years?

What needs to be my account balance today to be able to purchase the payout (annuity) contract?

(A.5) Payout Phase

What needs to be the initial capital $K_0 = PV$ if there should be an annuity of 24,000 EUR over a time horizon of 20 years. Please consider an interest rate of 3% and annuity-immediate.

$$(F): PV = \frac{FV}{q^N} = \frac{1}{q^N} * \frac{q^N - 1}{q - 1} * C = \frac{q^N - 1}{q^{N+1} - q^N} * C = C * PVF$$

$$(1): PV = \frac{1.03^{20} - 1}{1.03^{21} - 1.03^{20}} * 24,000 \text{ EUR} = \mathbf{357,059.40 \text{ EUR}}$$

$$(2): FV_{Savings} = 357,059.40 \text{ EUR} = C_{Saving} * \frac{q^N - 1}{q - 1}$$

$$(3): \text{Assumption: } r = 0.03, N = 40 \text{ years}$$

$$(4): C_{Savings} = FV_{Savings} * \frac{q - 1}{q^N - 1} = 357,059.40 \text{ EUR} * \frac{0.03}{1.03^{40} - 1} = \mathbf{4,735.46 \text{ EUR}}$$

(A.6) Payout Phase

How long can a person receive an annual amount of 24,000 EUR (annuity-immediate) if we consider an initial capital of 500,000 EUR and the interest rate equals 3%.

$$(F_1): PV = \frac{FV}{q^N} = \frac{1}{q^N} * C * \frac{q^N - 1}{q - 1}$$

$$(F_2): PV * q^N * (q - 1) = C * q^N - C$$

$$(F_3): C * q^N - PV * q^N * (q - 1) = C$$

$$(1): q^N * (C - PV * (q - 1)) = C$$

$$(2): N * \ln(q) = \ln(C) - \ln(C - PV * (q - 1))$$

$$(3): N = \frac{\ln(C) - \ln(C - PV * (q - 1))}{\ln(q)}$$

$$(4): N = \frac{\ln(24,000) - \ln(24,000 - 500,000 * (0.03))}{\ln(1.03)} = 33.18 \text{ years}$$

(A.7) Payout Phase

Mr. Huber wants to sell his house for 500,000 EUR. The buyer does not pay with cash but will pay an annuity over the next 20 years. What will be the annuity-immediate if we assume an interest rate of 7%?

$$A = 22,831.23 \text{ EUR}$$

$$B = 38,476.25 \text{ EUR}$$

$$C = 47,196.46 \text{ EUR}$$

$$D = 54,612.12 \text{ EUR}$$

$$(F): C = PV * \frac{q^{N+1} - q^N}{q^N - 1}$$

$$(1): C = 500,000 \text{ EUR} * \frac{1.07^{21} - 1.07^{20}}{1.07^{20} - 1} = 47,196.46 \text{ EUR}$$

(A.8) Payout Phase

Mr. Huber gets paid his life insurance of 250,000 EUR.

(a) How long can he receive an annuity-immediate of 24,000 EUR if the interest rate equals 3.25%?

(b) How long can he receive an annuity-due of 24,000 EUR if the interest rate equals 3.25%?

A = 13y both

B = (a) 13y, (b) 14y

C = (a) 14y, (b) 13y

D = 14y both

$$(1): N = \frac{-\ln\left(1 - \frac{PV * r}{C}\right)}{\ln(q)}$$

$$(2): N = \frac{-\ln\left(1 - \frac{250,000 \text{ EUR} * 0.0325}{24,000 \text{ EUR}}\right)}{\ln(1.0325)} = 12.92 \text{ years}$$

$$(3): N_{due} = \frac{-\ln\left(q - \frac{PV_{due} * r}{C_{due}}\right)}{\ln(q)} + 1$$

$$(4): N_{due} = \frac{-\ln\left(1.0325 - \frac{250,000 \text{ EUR} * 0.0325}{24,000 \text{ EUR}}\right)}{\ln(1.0325)} + 1 = 12.42 \text{ years}$$